MIL-T-14091D(ME)
23 November 1981
SUPERSEDING
MIL-T-14091C(ME)
6 December 1976

MILITARY SPECIFICATION

TRIPOD, SURVEYING; EXTENSION-LEG-TYPE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an extension-leg-type tripod for surveying instruments.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/2	- Screw-Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS15747	- Plumb Bobs, Brass, Plain or Quick Precision Adjustment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2016 - Moisture Content of Wood.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The tripod shall be the extension-leg type and shall be equipped with a threaded instrument fastener, triangular shape head, a headcap, a carrying sling, and a case attached to the tripod containing a wrench, plumb bob, cord, cord adjuster and cord centering instrument fastener fixture.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.3).

3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (6.4).

3.3.1 Metal parts. All metal components of the tripod shall be nonmagnetic, except for the corrosion resistant steel (CRS) points.

3.3.2 Webbing. Cotton webbing shall conform to MIL-W-530, type IIa, class 4, color olive drab.

3.3.3 Wood. One of the following species of hardwood shall be used: White ash, European beech, yellow birch, pecan hickory, or hard maple. Slope of grain shall not exceed 1 inch in 15 inches. Moisture content shall be not less than 7 percent or more than 15 percent when determined in accordance with ASTM D 2016.

3.4 Screw threads. All screws and screw threads shall be in accordance with FED-STD-H28.

3.5 Shock. The tripod, extended to its full length, shall not be damaged when dropped on a flat wood surface.

3.6 Stability. The tripod, set up for operation, shall withstand a superimposed load of not less than 100 pounds without slippage of the joints, abrupt bends of the legs at the joints, or abrupt bending in the joints. There shall be no permanant deformation of the legs, and each leg shall resume its original position upon removal of the load. Provisions shall be made to tighten all tripod connections to accomodate shrinkage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

3.7 Tripod head. The tripod head (see figure 1) shall be approximately triangular shaped; each side shall be 6.75 inches long plus or minus 0.75 inch. A 2.25 to 2.50-inch diameter hole shall be provided in the tripod head; the center of the hole shall be located at the bisectors of the angles formed by the sides of the tripod head. Bearing blocks shall be provided at each vertex for hinge, pin-type bearings, not less than 0.375 inch long and 0.375 inch in diameter. A single stud bolt and a fiber washer shall be located under the tripod head at each vertex to apply pressure on the bearing blocks for the adjustment of tension on the hinges.

3.7.1 Instrument fastener. An instrument fastener (see figure 1) shall hold the surveying instrument in place on top of the tripod head. The fastener shall be attached to the tripod head by a sliding arrangement. The sliding arrangement shall be pivoted and fastened at one end to the bottom of the tripod head allowing the instrument to be positioned anywhere within the perimeter of the hole in the center of the tripod head. Threads on the fastener shall be 0.625-11 UNC-2A conforming to FED-STD-H28/2. When the instrument fastener is pushed upward to its extremity it shall protrude above the tripod head not less than 0.375 inch and not more than 0.50 inch. When not in use, the instrument fastener shall drop below the surface of the head. Provision shall be made for suspending a plumb bob from the instrument fastener center in such a way that the pivot point of the plumb bob line itself or of the plumb bob line extended by the plumbline device shall be within 0.50 inch, plus or minus 0.0625 inch, of the top of the tripod head when fastened to an instrument. The total unobstructed plumbline swing shall be not less than plus or minus 5 degrees from the vertical. The plumbline holding device shall be a quick-connect/disconnect type so that the plumb bob can be removed for use of the optical plumb. The grip portion of the instrument fastener shall be long enough to permit the fastening of an instrument by the operator wearing arctic gloves.

3.8 Hinges. The hinges shall provide a firm connection between the legs and the tripod head. Pin-type bearings of the hinge not less than 0.375 inch long and not less than 0.375 inch in diameter shall connect the hinge to the bearing blocks of the tripod head.

3.9 Legs. The three tripod legs shall be the extension-leg type, enabling the lower part of the leg to slide into the upper part of the leg. When the leg is extended to its full length, the lower part of the leg shall lap not less than 2.5 inches into the upper part of the leg. The extended length, measured from the center of the hinge to tip of the steel point shall be 70 inches, plus or minus 2 inches. The closed length of the leg, measured from the center of the hinge to the tip of the steel point, shall be 44 inches, plus or minus 2 inches. A cotton webbing or leather strap with buckle shall be attached to the outside of one leg to hold the three legs together.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

3.9.1 Upper part. The upper part of the tripod leg shall consist of two wooden dowels or two aluminum dowels, a metal spacer, and a metal band with a clamping device for holding the lower part of the leg in place when adjusting the tripod to the required height.

3.9.2 Lower part. The lower part of the tripod leg shall consist of two wooden dowels or two aluminum dowels, or a single oblong section made from aluminum or wood, two metal stops, a metal foot spur, and a metal shoe with a steel point.

3.9.2.1 Metal shoe. The metal shoe shall be 4 to 7 inches long and a thickness and width to form a socket-type arrangement to accommodate the lower section. The lower end shall be seated in the socket of the metal shoe. The lower end of the metal shoe shall be tapered.

3.9.2.2 Steel point. The top cross section of the steel point shall match the bottom cross section of the metal shoe as to dimensions and configuration. The point shall be fastened to the metal shoe except when a replacable point is supplied, the top of the point may have a hex-shaped cross section of proper dimensions so that the wrench supplied (see 3.13) can be used to tighten or remove the replaceable points. There shall be a taper beginning with the taper on the metal shoe to the tip of the point. The length of the entire taper, including metal shoe and the point shall be 3 to 4 inches. The point shall be CRS and hardened to RC 40-50.

3.9.2.3 Foot spur. There shall be a foot spur on the outside of the metal shoe on each leg and shall extend out from the face of the shoe 1.50 inches plus or minus 0.50 inch.

3.9.3 Construction. When wood is used in the construction of the legs, all joints shall be secured with oval-head wood screws countersunk into the metal part of the joint. At joints that require tightening, hex head lag screws or self-locking hex head bolts and nuts may be used. When aluminum dowels are used in the construction of the legs, the joints shall be secured with aluminum rivets, peened to a button head. All joints shall be seured with waterproof cement to prevent penetration of moisture.

3.10 Headcap. The headcap shall be nonmagnetic metal or high-impact plastic. Provision shall be made for fastening the headcap to one of the metal spacers and for holding the headcap flat against the tripod leg. The headcap shall have provisions to be attached to cover the tripod head with the instrument fastener. If high impact plastic is used as the headcap material, a nonmagnetic metal thread insert shall be molded in the headcap to mate with the instrument fastener.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

3.11 Battery box attachment. One of the metal spacers attached to the upper dowels of the tripod leg shall be used as a bracket for holding the battery box to the outside of the tripod leg. The bracket shall be offset 0.375 inch, plus or minus 0.0625 inch, from the tripod leg. The upper edge of the bracket shall be notched, 1.375 inches long by 0.125 inch deep, plus or minus 0.0625 inch.

3.12 Plumb bob. The plumb bob shall conform to MS15747-10. The plumb bob shall be furnished with not less than 7.50 feet of braided linen cord, cord adjuster and instrument fastener adaptor all assembled for operation.

3.13 Wrench. A wrench shall be furnished to permit the adjustment of all bolts and nuts.

3.14 Case. A case made from leather, cotton webbing, or high-impact plastic shall hold the wrench and plumb bob assembly. The case shall be attached to the inside of one of the tripod legs. The case shall not hinder the full movement of any part of the tripod leg.

3.15 Carrying sling. An adjustable, padded carrying sling of cotton webbing shall be fastened to one leg of the tripod. The sling shall be sufficient length for carrying the tripod on the shoulder.

3.16 Telescopic plumbing rod. When specified (see 6.2), a telescopic plumbing rod with a mounted circular level shall be furnished. The plumbing rod shall be nonmagnetic metal. The level shall have a sensitivity of not less than 20 minutes per 2 millimeters. The rod shall be graduated so that the height of the surveying instrument may be determined to the nearest 0.1 inch. The rod shall be attachable to the instrument fastener by a quick-release lock. Provision shall be made for firm attachment of the rod to one leg of the tripod. The rod shall be fabricated so that the inner and outer members shall move over the fully extended length without binding or bending.

3.17 Weight. The weight of the tripod excluding the plumb bob, case, wrench, and carrying sling, shall not exceed 15.50 pounds.

3.18 Identification marking. The tripod shall be identified in accordance with MIL-STD-130.

3.19 Treatment and painting. The portions of the tripod normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type A, color and gloss as specified (see 6.2). The instrument contact surface of the tripod head and the threaded portion of the instrument fastener shall not be painted.

Source: https://assist.dla.mil -- Downloaded: 202640-18T02:51Z
Check the source to verify that this is the currentversion before use.

3.2 Workmanship. All edges shall be rounded or beveled. All joints shall be firm and dowels shall be firmly seated in their sockets. Dowels shall be straight. Metal and wood shall be free from cracks, scratches or splits. Screws and bolt-holes shall be drilled and shall have burrs removed. Bolts, nuts, screws, and rivets shall be tight. There shall be no binding when shortening or lengthening the tripod legs. The tripod shall not wobble and legs shall not bend when the tripod is extended to its full length. Paint shall be evenly applied on all painted surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article tripod shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article tripod shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects as specified in 4.5.1. AQL shall be 0.65 percent defective for major defects and 1.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 0.65 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tripods shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Screwthreads not as specified.
103. Metal parts missing or not as specified.
104. Cotton webbing not as specified.
105. Tripod head not as specified.
106. Instrument fastener not as specified.
107. Hinges not as specified.
108. Plumb bob assembly not as specified.
109. Tripod legs not as specified.
110. Metal shoe not as specified.
111. Steel point not as specified.
112. Foot spur not as specified.
113. Construction not as specified.
114. Headcap not as specified.
115. Battery box attachment not as specified.
116. Telescopic plumbing rod not as specified.

Minor

201. Wrench not as specified.
202. Case not as specified.
203. Carrying sling not as specified.
204. Weight not as specified.
205. Identification marking missing, incomplete, or illegible.
206. Treatment and painting not as specified.
207. Workmanship not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

4.5.2 Tests.

4.5.2.1 Shock to its full length with protection cap in place, shall be ition not less than 16 inches above a flat wood surface, drall from the vertical to a horizontal position of the wooshall be performed three times, allowing the tripod to fall opresence of broken, cracked, bent, or loose parts shall cons test.

4.5.2.2 Stabil be set up in a position for operation on a flat wood or conaded to its full height with each leg forming not lesswith the vertical. A stop shall be provided to prevent the bfrom splaying out of place. The tripod head shall be level a for holding the lower part of the leg in place shall be has than a 100-pound load shall then be superimposed on the tf slippage, abrupt bends, or abrupt bending at any joint shaof this test. In addition, upon removing the load from thg deflection in the leg indicating permanent deformation shalf this test.

4.6 Inspectio

4.6.1 Quality of pack.

4.6.1.1 Unit urpose of inspection, a completed pack prepared for shipmea unit of product.

4.6.1.2 Samplnination shall be in accordance with MIL-STD-105.

4.6.1.3 Examined in accordance with 4.6.1.2 shall be examined for the)L shall be 2.5 percent defective.

Major

117. Mate
118. Conta
119. Methc
120. Headc tripod head.
121. Plumbntained in the case.
122. Plumb not attached to leg as specified.
123. Legs ith strap.
124. Wood the end of box as specified.
125. Box c specified.
126. Strap
127. Markit, incomplete, or missing.

s shall be be straight. and bolt- ews, and engthening when the on all painted

e contract, quirements ne con- rmance of e Government. : forth in e supplies

.ble for and tested ole.

ified here-

ecified in

in 4.5.2.

rdance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each complete tripod shall be preserved in accordance with MIL-P-116, method III using a box conforming to PPP-B-636, W5c, style optional. The headcap shall be secured in place on the tripod head. The plumb bob and wrench shall be contained in the case. The plumbing rod, when specified (see 3.16), shall be attached to the leg, and the legs shall be secured together with the strap. A fitted wood block shall be inserted in the end of the container to prevent damage to the container from the points. Box closure shall be in accordance with the appendix to the box specification method V.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The tripods shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class I, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. Packing shall be as specified for level A, except that boxes shall be domestic class or type, and strapping shall be finish A.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tripod is intended for use with theodolites, levels, electronic distance measuring devices, and other surveying instruments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and number of units required (see 3.2).

c. When a plumbing rod is required (see 3.16).

d. Color and gloss required (see 3.19).

e. Degree of preservation and degree of packing required (see 5.1 and 5.2).

f. When other than finish B strapping is required (see 5.2.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first produced tripod. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Review activity:
DLA - GS

User activity:
Army - AR

Preparing activity:
Army - ME

Project 6675-A236

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

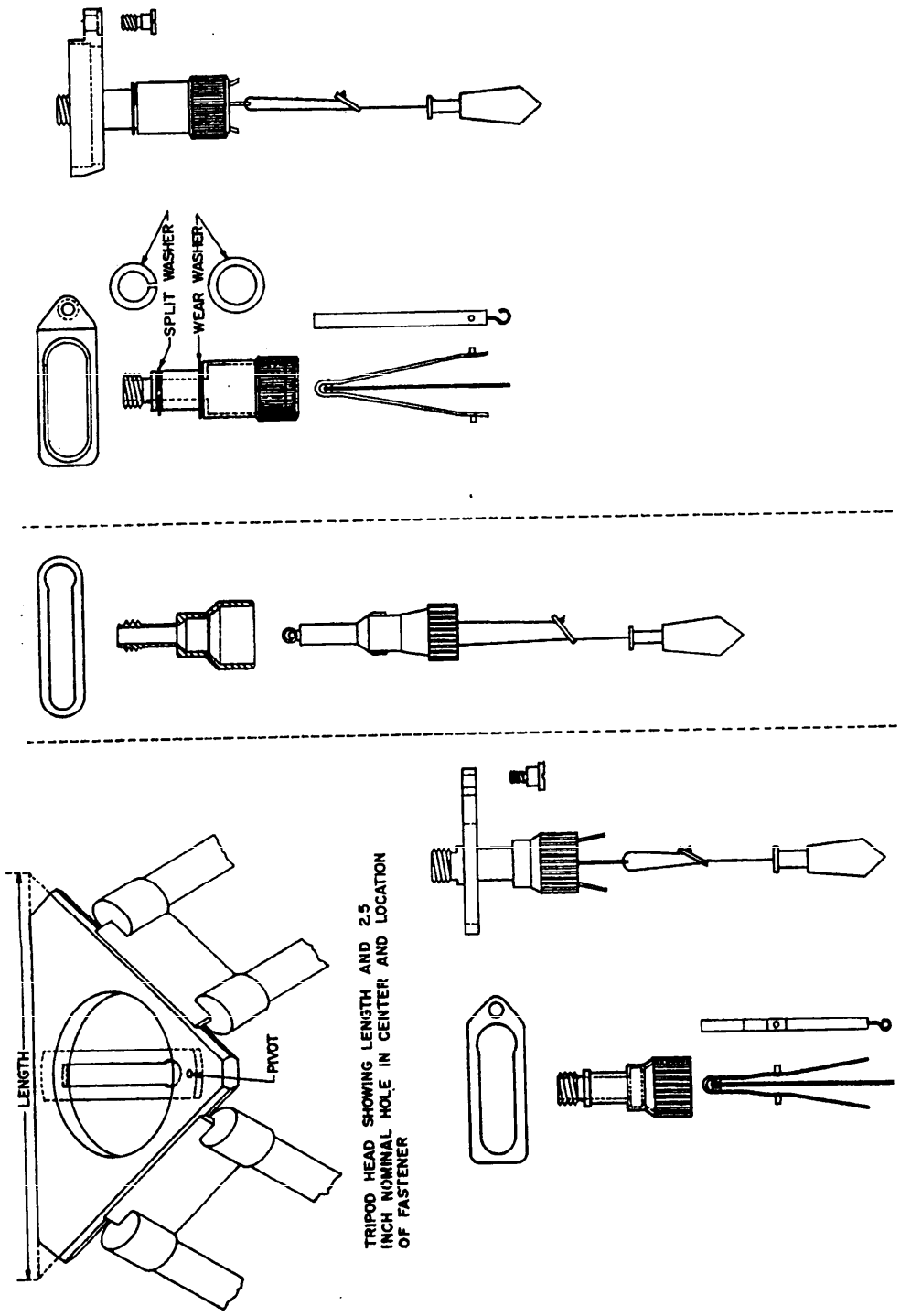


FIGURE 1. Instrument fasteners tripod head.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-14091D(ME) Tripod, Surveying; Extension-Leg-Type

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:51Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:51Z
Check the source to verify that this is the current version before use.